Beta Function Calculator

Deliverable 3: Numerical Correctness, Code Quality, Accessibility, and Testing

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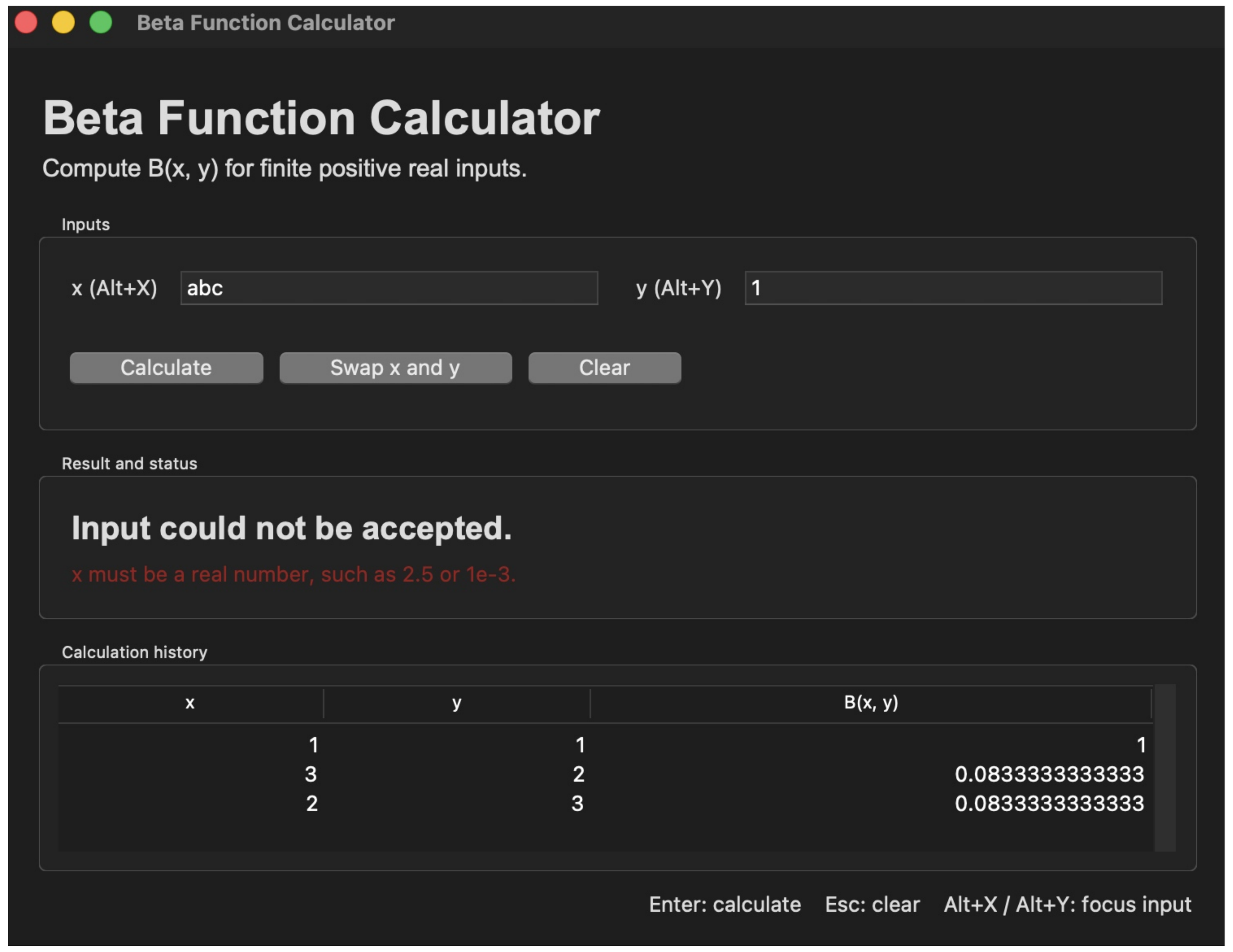
Semantic Version 1.0.0

$$B(x,y)=\frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)},\qquad x>0,\ y>0$$

Final Graphical Interface



The live interface shows a field-specific nonnumeric-input message while preserving earlier successful calculations. The user can correct the input and continue; native Tkinter widgets retain familiar focus and tab behavior.

Accessibility and Recovery

Keyboard access
Enter, Esc, Alt+X, Alt+Y

Recognition over recall
visible labels and actions

Helpful recovery
field-specific text messages

Status is not color-only
words identify success or error

The GUI aims to be accessible through readable labels, visible focus, redundant text feedback, keyboard operation, and recoverable validation. These choices align the interface with the user's task model and established usability heuristics (Kamthan, 2026c; Nielsen Norman Group, 2026).

Exception policy: expected input and numeric-range errors receive specific messages; unexpected arithmetic failures are not hidden behind a generic message and remain available for diagnosis.

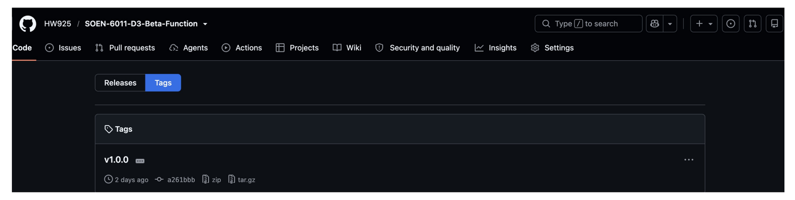
D3 conclusion: the release improves numerical stability, accessibility, and software quality through a stable log-beta implementation, explicit range behavior, and repeatable verification.

Problem 7: Quality and Versioning

```
Flake8 passed: no violations found.
```

```
------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

Correction made: Pylint reported `too-many-public-methods` after test expansion. One oversized test class was split into three focused classes, restoring **10.00/10** without a global suppression.



Why 1.0.0: first stable GUI and numerical contract, recorded by repository tag `v1.0.0` (Semantic Versioning, 2026).

Problem 7: Debugger Evidence

```
(Pdb) break beta_core.beta_function
Breakpoint 1 at /Users/weihuang/Documents/SOEN 6011/D3_F6_Beta_Function/beta_core.py:256
(Pdb) continue
> /Users/weihuang/Documents/SOEN 6011/D3_F6_Beta_Function/beta_core.py(263)beta_function()
-> if not is_finite(x_value):
(Pdb) p (x_value, y_value)
(2.0, 3.0)
(Pdb) next
> /Users/weihuang/Documents/SOEN 6011/D3_F6_Beta_Function/beta_core.py(265)beta_function()
-> if not is_finite(y_value):
(Pdb) next
> /Users/weihuang/Documents/SOEN 6011/D3_F6_Beta_Function/beta_core.py(267)beta_function()
-> if x_value <= 0.0:
(Pdb) next
> /Users/weihuang/Documents/SOEN 6011/D3_F6_Beta_Function/beta_core.py(269)beta_function()
-> if y_value <= 0.0:
(Pdb) continue
B(2, 3) = 0.08333333333333313
The program finished and will be restarted
> /Users/weihuang/Documents/SOEN 6011/D3_F6_Beta_Function/debug_beta.py(1)<module>()
-> """Small driver used only to demonstrate pdb on the Beta calculation."""
(Pdb)
```

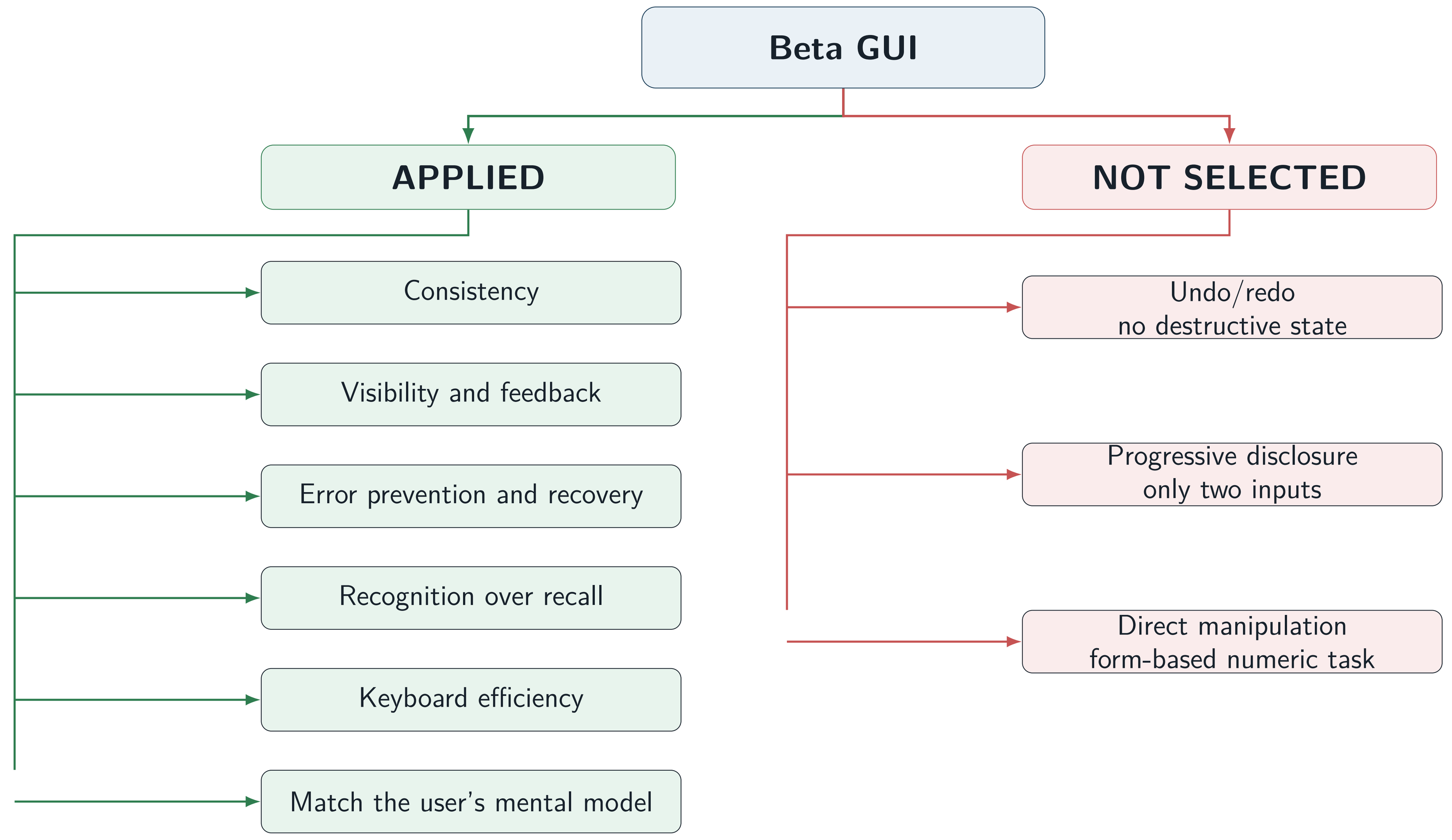
A reproducible pdb session stops inside `beta_function`, inspects $(x,y)=(2,3)$, and steps through finite and domain validation before continuing to the known result $1/12$.

Verified Numerical Contract

```
External verification only - mpmath, 100 decimal digits
Seed: 6011; generated cases: 500
Inputs: log-uniform from 1e-12 through 1e12
Normal representable results checked: 414
Range-limited reference results excluded: 86
Failures above 1e-10 relative error: 0
Maximum relative error: 1.223e-11
Worst normal-result input: (131117.64197605842, 81.32262207870075)
Large regression B(1e100, 1) relative error: 5.419e-13
Roundable boundary exp(-745): 4.9406564584124654e-324
```

Contract: accepted inputs return a nonzero finite binary64 result when representable; otherwise the program reports overflow or underflow. Normal results in the deterministic oracle set meet 10^{-10} relative error; subnormals follow binary64 spacing (NIST DLMF, 2026; mpmath, 2023).

UIDP Decision Mind Map



The mind map records both selected and rejected principles instead of treating every principle as mandatory (Kamthan, 2026d; Nielsen Norman Group, 2026).

Problem 8: PyUnit Test Suite

```
test_half_half_equals_pi (test_beta_core.BetaFunctionValueTests.test_half_half_equals_pi)
B(0.5, 0.5) equals pi. ... ok
test_identity_one_y (test_beta_core.BetaFunctionValueTests.test_identity_one_y)
B(1, y) equals 1/y. ... ok
test_identity_x_one (test_beta_core.BetaFunctionValueTests.test_identity_x_one)
B(x, 1) equals 1/x. ... ok
test_known_value_one_one (test_beta_core.BetaFunctionValueTests.test_known_value_one_one)
B(1, 1) equals one. ... ok
test_known_value_two_three (test_beta_core.BetaFunctionValueTests.test_known_value_two_three)
B(2, 3) equals one twelfth. ... ok
test_near_maximum_asymmetric_identity (test_beta_core.BetaFunctionValueTests.test_near_maximum_asymmetric_identity)
A near-maximum finite x does not trigger log-gamma cancellation. ... ok
test_regression_large_asymmetric_identity (test_beta_core.BetaFunctionValueTests.test_regression_large_asymmetric_identity)
The D3 large-input cancellation defect remains corrected. ... ok
test_symmetry (test_beta_core.BetaFunctionValueTests.test_symmetry)
Swapping x and y does not change the result. ... ok
test_missing_input_is_rejected (test_beta_core.InterfaceAndInputTests.test_missing_input_is_rejected)
The GUI parser identifies a missing field before calculation. ... ok
test_negative_x_is_rejected (test_beta_core.InterfaceAndInputTests.test_negative_x_is_rejected)
A negative x is rejected with a field-specific error. ... ok
test_negative_y_is_rejected (test_beta_core.InterfaceAndInputTests.test_negative_y_is_rejected)
A negative y is rejected with a field-specific error. ... ok
test_nonfinite_input_is_rejected (test_beta_core.InterfaceAndInputTests.test_nonfinite_input_is_rejected)
```

```
NaN and positive or negative infinity are rejected. ... ok
test_nonnumeric_input_is_rejected (test_beta_core.InterfaceAndInputTests.test_nonnumeric_input_is_rejected)
The GUI parser reports a useful nonnumeric-input error. ... ok
test_overflow_is_reported (test_beta_core.InterfaceAndInputTests.test_overflow_is_reported)
A result above the largest binary64 value reports a range error. ... ok
test_scientific_notation_parser (test_beta_core.InterfaceAndInputTests.test_scientific_notation_parser)
The GUI parser accepts scientific notation. ... ok
test_underflow_is_reported (test_beta_core.InterfaceAndInputTests.test_underflow_is_reported)
An unrepresentable positive result is not returned as zero. ... ok
test_unexpected_arithmetic_error_is_not_hidden (test_beta_core.InterfaceAndInputTests.test_unexpected_arithmetic_error_is_not_hidden)
Unexpected arithmetic failures remain visible for diagnosis. ... ok
test_zero_is_rejected (test_beta_core.InterfaceAndInputTests.test_zero_is_rejected)
Zero is rejected because it is outside the Beta domain. ... ok
test_zero_y_is_rejected (test_beta_core.InterfaceAndInputTests.test_zero_y_is_rejected)
A zero y is rejected with a field-specific error. ... ok
test_roundable_subnormal_exponential (test_beta_core.NumericalBoundaryTests.test_roundable_subnormal_exponential)
A finite result near the subnormal boundary is not rejected. ... ok
test_unroundable_exponential_is_rejected (test_beta_core.NumericalBoundaryTests.test_unroundable_exponential_is_rejected)
An exponential below the rounding boundary reports underflow. ... ok

----------------------------------------------------------------------
Ran 21 tests in 0.002s

OK
```

21/21 tests pass. Coverage includes trusted values, $B(1,y)=1/y$, $B(x,1)=1/x$, symmetry, the D3 large-input regression, separate x/y domain errors, NaN/infinity, overflow, underflow, and both sides of the subnormal rounding boundary.

Generative AI and CASTROFF. Tool: OpenAI ChatGPT/Codex. **Constraints:** preserve from-scratch code and honest range claims; **Audience:** TA/instructor; **Structure:** evidence by Problems 7 and 8; **Tone:** concise and technical; **Role:** graduate student reviewer; **Output format:** code/test suggestions and poster statements; **Focus:** quality, accessibility, binary64 boundaries, and regression coverage; **Function:** identify missing evidence and unsupported claims. Example prompts requested a PEP 8 and range-behaviour review and focused PyUnit regressions. Suggestions were executed and revised; unverified claims were rejected. Full records are in `GAI_USAGE.md`.

References. Kamthan, P. (2026a), *SOEN 6011 Project Description*; Kamthan, P. (2026b), *Building, Versioning, and Releasing Software*; Kamthan, P. (2026c), *Introduction to Mental Models*; Kamthan, P. (2026d), *Brainstorming and Mind Mapping*; Nielsen Norman Group (2026), *Ten Usability Heuristics*; Semantic Versioning (2026), *Semantic Versioning 2.0.0*; NIST DLMF (2026), *Section 5.12: Beta Function*; mpmath (2023), *mpmath 1.3.0 Documentation*; Van Rossum, Warsaw, and Coghlan (2001), *PEP 8*; Python Software Foundation (2026), *unittest and pdb Documentation*.

Repository: github.com/HW925/SOEN-6011-D3-Beta-Function